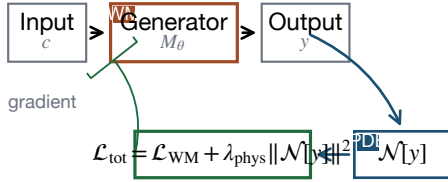


# Four mechanisms for integrating physical constraints into world models

## a Constraint injection (soft)



Soft constraint:  $\|\mathcal{N}[y]\|$  minimised but not bounded.

## b Architecture embedding (hard)

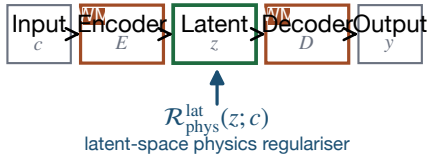


$$\forall z: \|\mathcal{N}[\mathcal{T}_{\text{phys}}(z)]\| \leq \epsilon_1$$

hard physical guarantee — bound holds for every input

Examples: data-consistency layer [37], Learned Primal-Dual [12], physics-driven diffusion [60], physics-encoded transformer [215]

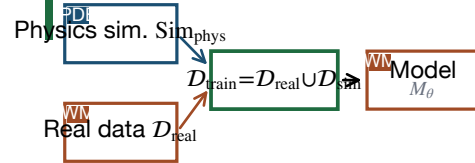
## c Latent-space regulation



latent-space physics regulariser

ReSample [90]: hard data consistency in latent space

## d Simulation-in-the-loop



$$D_{\text{sim}} = \{(c, \text{Sim}_{\text{phys}}(c) + \eta)\}$$

100x data efficiency reported on rare anatomies [233]

Fig.\,2. Block-level architectures of the four PIWM integration mechanisms. Tags:  $\text{WM}$  = world-model component,  $\text{PDE}$  = physics component.